

Student Performance Analysis

Exploratory Data Analysis (EDA) Report

Objective

To explore the student performance dataset, assess data quality, summarize statistical properties, identify outliers, analyze relationships among variables, and derive insights from visualizations.

Dataset Exploration

The notebook inspects the dataset using shape, column names, data types, descriptive statistics, and missing-value checks before analysis.

Data Cleaning

Missing values are checked. Numerical outliers are detected and treated using the Interquartile Range (IQR) method to improve analysis quality.

Statistical Analysis

The notebook computes mean, median, mode, variance, and standard deviation for numerical features and generates a correlation matrix to study relationships.

Visualizations

Visualizations include a gender distribution bar chart and a line chart of final grades (G3), supporting interpretation of dataset characteristics and performance trends.

Key Insights

- Initial exploration confirms the dataset is suitable for analysis after quality checks.
- Descriptive statistics summarize the spread and central tendency of numerical variables.
- IQR-based outlier treatment reduces the influence of extreme values.
- Correlation analysis highlights relationships among academic variables for future predictive modeling.
- Visualizations improve understanding of gender distribution and final-grade patterns.
- The cleaned dataset is appropriate for further machine learning or predictive analytics.

Conclusion

The exploratory data analysis provides a structured understanding of the dataset and prepares it for advanced statistical analysis and machine learning tasks.